

Shutdown Unit Indicates Shut Down Cause When Shutdown Did Not Occur

Symptoms

The engine does **not** shut down, but the SDU410 unit falsely indicates that a shutdown has occurred.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Check that the Shutdown Override is **not** active. If the Shutdown Override is active and no shutdown occurred but an alarm informs operator shut down would have occurred if it had **not** been overridden. Deactivate the shutdown override alarm. If the alarm can **not** be deactivated, reference the appropriate troubleshooting tree.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box	
	STEP 1A. Check for alarm and LED illumination.	LED flashing?
	STEP 1A-1. Check the shutdown unit Modicon™ communication bus supply and return wires for an open.	Less than 10 ohms?
	STEP 1A-2. Check the shutdown unit Modicon™ communication bus supply and return wires for a wire-to-wire short.	Less than 10 ohms?

STEPS		SPECIFICATIONS
	STEP 1A-3. Check the shutdown unit Modicon™ communication bus supply wire for a short to ground.	Less than 10 ohms?
	STEP 1A-4. Check the engine protection override signal and return wires for an open.	Less than 10 ohms?
	STEP 1A-5. Check the engine protection override signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 1A-6. Check the engine protection override signal wire for a short to ground.	Less than 10 ohms?
	STEP 1A-7. Check the engine protection override relay supply and signal wires for an open.	Less than 10 ohms?
	STEP 1A-8. Check the engine protection override relay signal and return wires for wire-to-wire short.	Less than 10 ohms?
	STEP 1A-9. Check the engine protection override relay signal wire for a short to ground.	Less than 10 ohms?

STEP 1. Check the customer interface box.

STEP 1A. Check for alarm and LED illumination.

Conditions :

- Check for alarm and LED illumination.

Action	Specification/Repair	Next Step

Check the Shutdown Override circuit for activation.	LED flashing? YES Repair : Replace the SDU410 unit. Contact a Cummins® Authorized Repair Location.	Repair complete
	LED flashing? NO	1A-1

STEP 1A-1. Check the shutdown unit Modicon™ communication bus supply and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the shutdown unit Modicon™ communication bus supply and return wires at the SDU410 unit and DCU410 unit.

Action	Specification/Repair	Next Step
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Check the shutdown unit Modicon™ communication bus supply and return wires for an open. - Place one test lead on the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit. - Place the other test lead on the shutdown unit Modicon™ communication bus supply wire at the DCU410 unit. - Place one test lead on the shutdown unit Modicon™ communication bus return wire at the SDU410 unit. - Place the other test lead on the shutdown unit Modicon™ communication bus return wire at the DCU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1A-2
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 1A-2. Check the shutdown unit Modicon™ communication bus supply and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the shutdown unit Modicon™ communication bus supply and return wires at the SDU410 unit and DCU410 unit.

Action	Specification/Repair	Next Step
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Check the shutdown unit Modicon™ communication bus supply and return wires for a wire-to-wire short. • Place one test lead on the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit. • Place the other test lead on all other wires at the DCU410 unit. • Place one test lead on the shutdown unit Modicon™ communication bus return wire at the SDU410 unit. • Place the other test lead on all other wires at the DCU410 unit.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? NO	1A-3

STEP 1A-3. Check the shutdown unit Modicon™ communication bus supply wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit and DCU410 unit.

Action	Specification/Repair	Next Step
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Check the shutdown unit Modicon™ communication bus supply wire for a short to ground. - Place one test lead on the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	1A-4

STEP 1A-4. Check the engine protection override signal and return wires for an open.

Conditions : - Open the customer interface box. - Disconnect the engine protection override signal and return wires at the SDU410 unit and engine protection override relay.		
Action	Specification/Repair	Next Step

Check the engine protection override signal and return wires for an open. - Place one test lead on the engine protection override signal wire at the SDU410 unit. - Place the other test lead on the engine protection override signal wire at the engine protection override relay contact. - Place one test lead on the engine protection override return wire at the SDU410 unit. - Place the other test lead on the engine protection override return wire at the engine protection override relay contact. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1A-5
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 1A-5. Check the engine protection override signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the engine protection override signal and return wires at the SDU410 unit and engine protection override relay.

Action	Specification/Repair	Next Step
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Check the engine protection override signal and return wires for a wire-to-wire short. • Place one test lead on the engine protection override signal wire at the SDU410 unit. • Place the other test lead on all other wires at the SDU410 unit. • Place one test lead on the engine protection override return wire at the SDU410 unit. • Place the other test lead on all other contacts on the engine protection override relay. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	1A-6

STEP 1A-6. Check the engine protection override signal wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the engine protection override signal wire at the DCU410 unit.

Action	Specification/Repair	Next Step

Check the engine protection override signal wire for a short to ground. - Place one test lead on the engine protection override signal wire at the SDU410 terminal strip. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	1A-7

STEP 1A-7. Check the engine protection override relay supply and signal wires for an open.

Conditions : - Open the customer interface box. - Disconnect the engine protection override relay supply and signal wires at the engine protection override relay.		
Action	Specification/Repair	Next Step

Check the engine protection override relay supply and signal wires for an open. • Place one test lead on the engine protection override relay signal wire at the engine protection override relay. • Place the other test lead on the engine protection override relay signal pin at the C3 connector. • Place one test lead on the engine protection override relay signal wire at the engine protection override relay. • Place the other test lead on the engine protection override relay signal pin at the X4 connector. • Place one test lead on the engine protection override return wire at the SDU410 unit. • Place the other test lead on the engine protection override relay return wire at the engine protection override relay contact. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	1A-8 Repair complete
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STEP 1A-8. Check the engine protection override relay signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the engine protection override relay signal and return wires at the SDU410 unit and engine protection override relay.

Action	Specification/Repair	Next Step
Check the engine protection override relay signal and return wires for a wire-to-wire short. • Place one test lead on the engine protection override relay signal wire at the SDU410 unit. • Place the other test lead on all other wires at the SDU410 unit. • Place one test lead on the engine protection override relay return wire at the SDU410 unit. • Place the other test lead on all other contacts on the engine protection override relay.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? NO	1A-9

STEP 1A-9. Check the engine protection override relay signal wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the engine protection override relay signal wire at the DCU410 unit.

Action	Specification/Repair	Next Step
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Check the engine protection override relay signal wire for a short to ground. - Place one test lead on the engine protection override relay signal wire at the SDU410 terminal strip. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	Contact a Cummins® Authorized Repair Location